Wazuh Home SOC Lab

Technical & Investigation Report

Prepared by Dhruv Vats · GitHub: dhruvvats17

Wazuh 4.14.7 · Ubuntu Server · Windows 11 · Sysmon · Kali Linux

Report purpose	Document lab architecture, screenshot-supported findings, investigation workflow, limitations, and recommendations.
Evidence status	Based on owner-supplied screenshots and owner-reported test history. Not a production audit.
Evidence period	Screenshots supplied in August 2026; all dashboard counters are historical snapshots.

Public portfolio document · Private IP addresses omitted

1. Executive Summary

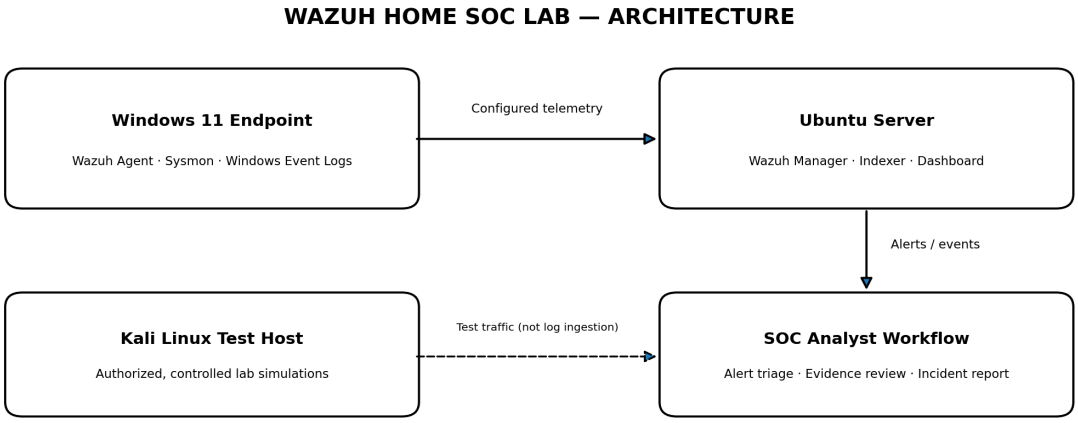
This project documents a small home Security Operations Center (SOC) lab built to practice endpoint telemetry collection, centralized security monitoring, alert triage, and incident documentation using Wazuh. A Windows 11 endpoint forwards telemetry to a Wazuh manager hosted on Ubuntu Server. Sysmon supplies detailed process and host-activity events. Kali Linux is used as a controlled testing machine for authorized lab-only activity.

The supplied screenshots show a Wazuh dashboard, an active Windows agent, Threat Hunting event views, repeated Windows authentication-failure events, a custom network-scan-related alert, and Windows Firewall log output. The network-scan screenshot displays “Nmap Scan Detected” with rule ID 100100 and level 10. This is evidence of an alert match in the lab, not proof of impact or malicious intent.

2. Objectives

• Collect Windows endpoint events centrally in Wazuh.
• Use Sysmon and Windows event telemetry to support investigation.
• Review security events in Wazuh dashboard and Threat Hunting.
• Demonstrate and document a network-scan-related alert.
• Practice SOC L1 triage and clear incident reporting.

3. Lab Architecture



Conceptual architecture. Confirm actual log sources, routing, and connectivity in the live lab.

Component	Role
Ubuntu Server	Hosts Wazuh manager, indexer, and dashboard components.
Wazuh 4.14.7	Centralized event analysis and dashboard.
Windows 11	Monitored endpoint with Wazuh Agent, Sysmon, Windows Event Logs.
Kali Linux	Authorized lab test source for network/security exercises.

Private home-network IP addresses are intentionally omitted from this public report.

4. Evidence Register

ID / file	What it supports	Limitation
E-01 Dashboard overview	Wazuh dashboard accessible; historical counters visible.	Point-in-time snapshot.
E-02 Agent status	Windows agent shown as active.	Does not prove current connectivity.
E-03 Threat Hunting	Event summary view was available.	Aggregates do not establish malicious activity.
E-04 Network scan alert	“Nmap Scan Detected”, rule 100100, level 10 shown.	Raw JSON and repeat test recommended.
E-05 Auth failures	Multiple Windows logon-failure events visible.	Cause requires event-level correlation.
E-06 Firewall log	Windows Firewall log output available.	Console lines do not prove Wazuh alert matching.

5. Findings and Analysis

Finding 1 — Endpoint telemetry path

The dashboard and agent-list screenshot indicate that the Windows endpoint was registered and shown as active at capture time. This supports the configured endpoint-to-manager monitoring path. Current status should be reconfirmed before a live demonstration.

Assessment: Configuration evidence present. **Next check:** Confirm current agent status and verify a fresh test event appears with a current timestamp.

Finding 2 — Network-scan-related alert

The Threat Hunting screenshot contains rows labelled “Nmap Scan Detected”, with rule ID 100100 and level 10. This supports that a network-scan-related rule matched events in the lab at capture time.

Assessment: Screenshot evidence of a matching alert is present. **Next check:** Export full event JSON, record test time/source host, repeat the authorized scan, and verify a new event.

Finding 3 — Windows authentication failures

The authentication-failure screenshot shows multiple logon-failure events. Repeated failures warrant triage but are not automatically proof of brute force. Inspect event IDs, usernames, source workstation/IP within the private lab, time range, frequency, and whether a successful logon followed the failures.

Assessment: Event evidence present; cause is not conclusively established by the screenshot alone. Correlate raw Windows Security events, commonly 4625 (failed logon) and 4624 (successful logon), with timestamps and source fields.

Finding 4 — Firewall logs

The Windows Firewall console screenshot shows firewall log entries and action/protocol fields. This supports that firewall logging was available for review, but it does not independently prove every entry was ingested or matched by Wazuh.

Assessment: Host-side log evidence present. **Next check:** Confirm collection configuration, generate one safe authorized test event, and verify the raw event and Wazuh rule.

Finding 5 — SSH brute-force test

The project owner reports that an SSH brute-force exercise and related alert were tested. No dedicated SSH alert screenshot or raw export was identified in the evidence reviewed, so this is recorded as author-reported rather than independently evidenced.

Assessment: Author-reported test; screenshot evidence pending. Add the alert details and a sanitized event export during a repeatable lab-only test.

6. SOC L1 Workflow

1. Validate the alert: record rule ID, severity, event time, agent, and description.
2. Scope the event: set the time range and identify endpoint/source.
3. Collect evidence: review alert JSON, Windows/Sysmon events, and firewall/authentication logs.
4. Correlate: compare repeated events, source/destination details, process/user fields, and nearby activity.
5. Assess severity based on confidence, asset importance, frequency, and potential impact.
6. Decide/escalate: classify only when evidence supports the conclusion.
7. Document findings, actions, limitations, and follow-up.
8. Do not claim containment or remediation unless performed and verified.

7. MITRE ATT&CK; Context

Potential mappings for investigation context (not proof of technique): **T1046 — Network Service Discovery** may be relevant to a network scan; **T1110 — Brute Force** may be relevant to repeated authentication failures when evidence supports password guessing; **T1078 — Valid Accounts** should be considered only when evidence shows suspicious successful use of a valid account.

8. Limitations

• Personal lab, not a production SOC.
• Screenshots are historical and do not prove current service health.
• Screenshots do not replace raw JSON/EVTX exports, timestamps, repeatability notes, or hashes.
• SSH brute-force is author-reported because dedicated screenshot evidence was not present.
• Authentication failures may be benign and need correlation.
• Firewall console output alone does not confirm end-to-end Wazuh rule matching.
• No containment action or real-world compromise is claimed.

9. Recommendations

1. Export sanitized JSON for the network-scan alert and authentication-failure events.
2. Repeat SSH and network-scan tests only inside the owned/authorized lab and capture timestamped screenshots.
3. Record test time, source/target lab roles, expected/actual outcome, and cleanup.
4. Capture before/after evidence showing the matching rule and event.
5. Revalidate custom rules and confirm end-to-end matching.
6. Redact private IPs, public IPs, usernames, tokens, and unrelated personal information.
7. Keep README claims aligned with evidence committed to the repository.

10. Conclusion

The available evidence demonstrates a Wazuh learning environment with a Windows endpoint shown as active, Threat Hunting event views, Windows authentication-failure telemetry, firewall logs, and a network-scan-related alert match. This report shows how a junior SOC analyst can move from alert validation to event correlation, evidence recording, and cautious written assessment. Raw-event exports and repeatable test captures will strengthen the portfolio.

Prepared by Dhruv Vats · https://github.com/dhruvvats17/wazuh-home-soc-lab